THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS
MATH3603 Probability theory

Tutorial 2 Solution

1. Bill and George go target shooting together. Both shoot at a target at the same time. Suppose Bill hits the target with probability 0.6, where as George, independently, hits the target with probability 0.3.

   (a) Given that exactly one shot hit the target, what is the probability that it's George's shot?

   (b) Given that the target is hit, what is the probability that George hit it?

   Solution:

   Let B be the event of Bill hitting the target, G br George hitting the target and S_i be exactly i shot(s) hit the target.

   (a) $P(G|S_1) = \dfrac{P(G \cap S_1)}{P(S_1)} = \dfrac{P(G \cap B^c)}{P(G \cap B^c) + P(B \cap G^c)} = \dfrac{0.3 \times 0.4}{0.3 \times 0.4 + 0.6 \times 0.7} = \dfrac{2}{9}.$

   (b) $P(G|S_1 \cup S_2) = \dfrac{P(G \cap (S_1 \cup S_2))}{P(S_1 \cup S_2)} = \dfrac{P(G)}{P(S_0^c)} = \dfrac{0.3}{1 - 0.4 \times 0.7} = \dfrac{5}{12}.$

2. Three prisoners A, B and C are informed by their jailer that one of them has been chosen at random to be executed, and the other two are to be freed. Prisoner A asks the jailer to tell him privately which of his fellow prisoners will be set free, claiming that there would be no harm in divulging this information, since he already knows that at least one will go free. The jailer refuses to answer this question, pointing out that if A knew which of his fellows were to be set free, then his own probability of being executed would rise from 1/3 to 1/2 since he would then be one of the prisoners. Is the jailer correct?

   Solution:

   This is basically the Monty Hall problem.

   Let $A/B/C$ be the event that prisoner A/B/C got executed and J_B/J_C be the event that the jailer says that prisoner B/C will be set free. Note that if B (respectively, C) will be executed, the jailer can only answer C (respectively, B); if A will be executed, the jailer can answer either B or C.

   Then $P(A|J_B) = \dfrac{P(A \cap J_B)}{P(J_B)} = \dfrac{P(A \cap J_B)}{P(A \cap J_B) + P(C \cap J_B)} = \dfrac{\frac{1}{3} \times \frac{1}{2}}{\frac{1}{3} \times \frac{1}{2} + \frac{1}{3}} = \dfrac{1}{3} = P(A).$

   Similarly $P(A|J_C) = P(A)$ as well. Thus the jailer is wrong.

3. A bowl contains twenty cherries, exactly fifteen of which have had their stones removed. A greedy pig eats five whole cherries, picked randomly, without any remarks on the presence or absence of the stones. Subsequently, a cherry is picked randomly from the remaining fifteen.

   (a) What is the probability that this cherry contains a stone?

(b) Given that this cherry contains a stone, what is the probability that the pig consumed at least one stone?

Solution:

(a) Let

$$\begin{aligned} A_i &= \{\text{pig eats } i \text{ cherries with stones}\}, \quad \forall i = 0, 1, \ldots, 5, \\ B &= \{\text{this cherry contains a stone}\}. \end{aligned} \tag{1}$$

Using the Law of total probability,

$$P(B) = \sum_{i=0}^{5} P(B|A_i)P(A_i)$$

where

$$P(B|A_i) = \frac{5-i}{15}, \text{ and } P(A_i) = \frac{\binom{5}{i}\binom{15}{5-i}}{\binom{20}{5}}$$

we get $P(B) = \frac{1}{4}$. Alternatively, one can view the problem as arrange these 20 cherries in a row, where the 6th cherry is to be picked. Therefore, the probability is simply $P(B) = \frac{5}{20} = \frac{1}{4}$.

(b) Let $A = \{\text{the pig consumed at least one stone}\}$, then

$$P(A^c|B) = \frac{P(A^c \cap B)}{P(B)}$$

where

$$P(A^c \cap B) = \frac{\binom{15}{5}\binom{5}{1}}{\binom{20}{5}\binom{15}{1}}, \text{ and } P(B) = \frac{1}{4}$$

Therefore,

$$P(A|B) = 1 - P(A^c|B) = 1 - \frac{15}{19} \cdot \frac{14}{18} \cdot \frac{13}{17} \cdot \frac{12}{16} \cdot \frac{11}{15}$$

4. The n passengers for a Bell-Air flight in an airplane with n seats have been told their seat numbers. They get on the plane one by one. The first person sits in the wrong seat. Subsequent passengers sit in their assigned seats whenever they find them available, or otherwise in a randomly chosen empty seat. What is the probability that the last passenger finds its seat free given that the first person sits in the wrong seat?

Solution: For those n passengers in the row, we denote them as $1, 2, \ldots, n$. First note that if the first passenger wrongly sits in the i_1-th passenger's seat, then the 2nd until the $(i_1 - 1)$-th passenger can find their own seat available, thus all sitting in the correct seat. While the i_1-th passenger will randomly choose another seat, for instance, the i_2-th passenger's seat. This procedure will be repeated until the i_k-th passenger choose the seat of the first passenger (note that $i_k \neq n$). In this case, $1 \to i_1 \to i_2 \to \cdots \to i_k \to 1$ is denoted as a closed walk. Now we let $P(k)$ be the probability that the first and exactly k other passengers in $\{2, \ldots, n-1\}$ form such a closed walk, then

$$P(k) = \sum_{2 \le i_1 < i_2 < \cdots < i_k \le n-1} \frac{1}{n} \cdot \frac{1}{n - i_1 + 1} \cdot \frac{1}{n - i_2 + 1} \cdots \frac{1}{n - i_k + 1}$$

Let $A = \{\text{the last passenger finds his seat free}\}$ and $B = \{\text{the first passenger sits in the wrong seat}\}$. Note that when A and B happens, the length k of any closed walk must be between 1 and $n-2$. We have $P(B) = \frac{n-1}{n}$ and

$$\begin{aligned}
P(A \cap B) &= \sum_{k=1}^{n-2} P(k) \\
&= \sum_{k=1}^{n-2} \sum_{2 \le i_1 < i_2 < \cdots < i_k \le n-1} \frac{1}{n} \cdot \frac{1}{n - i_1 + 1} \cdot \frac{1}{n - i_2 + 1} \cdots \frac{1}{n - i_k + 1} \\
&= \frac{1}{n} \cdot \left(\left(1 + \frac{1}{n-2+1}\right) \left(1 + \frac{1}{n-3+1}\right) \cdots \left(1 + \frac{1}{n-(n-1)+1}\right) - 1 \right) \\
&= \frac{n-2}{2n}
\end{aligned}$$

where in the third equation we use the fact that

$$\prod_{t=1}^{n} (1 + X_t) = 1 + \sum_{k=1}^{n} \sum_{1 \le i_1 < i_2 < \cdots < i_k \le n} X_{i_1} \cdots X_{i_k}$$

Hence,

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{n-2}{2(n-1)}$$